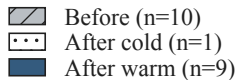
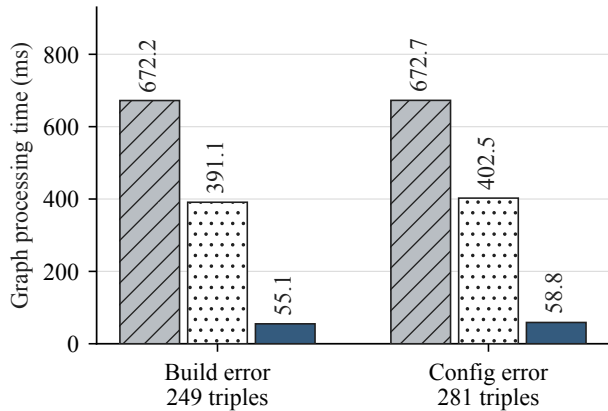


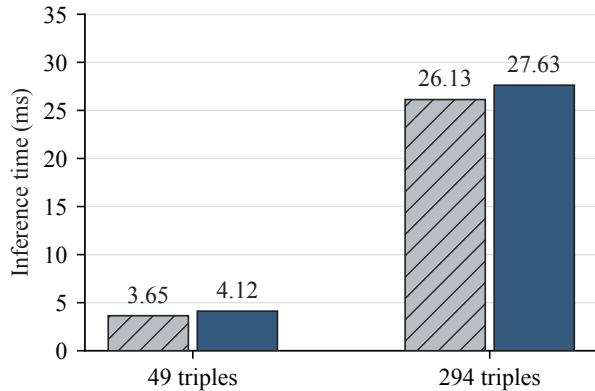
Local graph microbenchmarks; no model or network latency



(a) Shadow pipeline optimization



(b) Rule worker, 40 warm runs



(a) Cold includes process start; warm reuses a worker. (b) Graph sizes and revisions differ.  
These measurements are separate from the graph\_compact online pipeline.